

Work Experience

AI Engineer

Oct 2024 – Present

AXA GBS

Pune, India

- Developed a multi-agent system that converts natural language queries into structured SQL statements while returning tabular data or natural responses based on query context.
- Adaptive response generation, where the system decides if a query requires a text summary, table, and graphical visualization.
- Dynamic chart generation based on retrieved data, allowing responses in pie charts, bar charts, line graphs, or other formats, tailored to user questions.
- Enhanced analytics accessibility, enabling non-technical users to interact with databases via natural language and receive visually intuitive results.
- Supports real-time data updates, ensuring accurate representation of queried information through interactive reports and visual dashboards.
- Enhanced user accessibility by allowing non-technical users to request complex queries via text input, receive structured reports, and interact with real-time analytics.
- Delivered a proof-of-concept (PoC) for healthcare entities, leveraging Chain of Density (CoD) and few shot advanced prompting techniques to enhance summarization quality in medical records and reports.
- Implemented Retrieval-Augmented Generation (RAG) for company policy document processing, enhancing accuracy in AI-driven HR policy automation.
- Integrated Conversational Summary Buffer Memory in the RAG-based HR policy chatbot to maintain context across interactions and improve user experience.
- Evaluated AI-generated outputs using RAGAS framework alongside human assessment to ensure precision, compliance, and contextual relevance.
- Integrated human alignment methodologies within the RAGS framework, fine-tuning policy retrieval models to better reflect organizational needs.
- Used Large Language Models (LLMs) as judges alongside RAGAS for comparative evaluations, improving response quality and decision-making.
- Added Guardrails for safety, ensuring ethical and responsible AI-driven document retrieval, mitigating risks related to policy misinterpretation.
- Implemented validators to check AI-generated responses for toxicity, bias, and factual accuracy, ensuring reliable information output.
- Developed advanced pricing models for insurance using machine learning and Generative AI-based RAG systems. Engaged in client and stakeholder meetings, gathered insights, and delivered tailored solutions on schedule.

Jr. AI Developer

Oct 2022 – Oct 2024

Forage AI

Pune, India

- Researched and selected appropriate machine learning models based on data characteristics.
- Prepared high-quality datasets tailored for effective model training, ensuring alignment with project goals.
- Developed and fine-tuned deep learning models for diverse applications, including computer vision and natural language processing tasks.
- Integrated business clustering using Language Model (LLM) techniques to categorize and organize unstructured data effectively.
- Achieved a remarkable 95% accuracy in business clustering by combining K-means clustering algorithms with LLM capabilities (Llama-3, Mistral LLM), ensuring precise categorization of diverse business data.
- Leveraged LLM's contextual understanding to enhance clustering accuracy and provide nuanced insights into business data relationships.
- Utilized transfer learning and domain-specific knowledge to refine the Mistral model's ability to extract relevant information, classify data into specified categories, and derive actionable insights from financial documents.
- Designed and implemented the POC to demonstrate the effectiveness of GPT-4 vision API in various scenarios, including image recognition, object detection, and text extraction from documents.

- Led the hands-on fine-tuning of a Language Model (LLM) for Generative (Open Book) Question-Answering dataset using QLORA and PEFT in Python, improving its precision and relevance.
- Familiar with fine-tuning Llama2 and Falcon models as per business requirements.
- Finetuned Llama2 model with Instruction tuning that involves quantizing pretrained LLM to just 4 bits and small adaptors using just single GPU.
- Experience building scalable applications with LLMs, using frameworks such as LangChain.
- Hands-on experience with implementation of RAG techniques and Vector database as chroma DB with GPT-4 models for Proof of Concept (POC).
- Expertise in implementing Langchain QA retriever chains for private unstructured document question answering.
- Implemented Advanced RAG system using GPT-4 API for a healthcare expert search engine.
- Integrated sentiment analysis tools to gauge the sentiment expressed in the generated content, providing valuable insights into the emotional tone conveyed in the responses.
- Integrated Pinecone vector database for efficient information retrieval, ensuring a nuanced and tailored healthcare solution as per users' queries.
- Implemented document classification for financial documents using Deep-learning / Transformer model. Categorized content into specified categories based on client requirements, employing machine learning techniques for accurate classification.
- Collaborated closely with cross-functional teams to implement innovative solutions using transformer models, particularly for entity and information extraction tasks.
- Designed and implemented a robust parsing system for extracting crucial information from invoices.
- Proficient in unsupervised clustering of text data from website links, enabling meaningful insights and data organization.
- Conducted text summarization of website content within clustering projects for efficient data analysis.
- Utilized Python libraries to perform hands-on PDF-to-image conversion, enabling efficient text and image extraction for model training.
- Demonstrated proficiency in Optical Character Recognition (OCR) using tools like Tesseract to accurately extract text from various sources, including images and documents.
- Trained and deployed advanced object segmentation models using deep learning techniques, packaging them within Docker containers for streamlined integration and scalability in production environments.
- Engaged in continuous model training and retraining to drive consistent performance improvements.
- Implemented post-processing steps within the machine learning pipeline to ensure and maintain data quality throughout the process.
- Collaborated on team-wide initiatives to enhance model development practices, sharing insights and best practices.
- Integrated optimized models into scalable production workflows, enhancing overall system efficiency.

Data Engineer

Oct 2020 - Oct 2022

Neurapses

Remote, Pune

- Managed full project lifecycles, from designing and experimenting with models to deploying and maintaining production solutions. Played a key role in establishing CI/CD pipelines using Azure DevOps and Jenkins.
- Applied statistical and data mining techniques like boosting, generalized linear models/regression, random forests, and social network analysis to gain insights from data. Utilized data cleaning, analysis, and visualization methods.
- Successfully deployed a Recommendation Engine in production, which conditionally suggests menu items based on user history, contributing to increased average order size.
- Proficiently utilized and optimized Elastic search 7.0 indices to store and visualize data, including geo-location queries and mapping. Managed data ingestion into Elasticsearch using Logstash.
- Implemented Elasticsearch solutions, improving data storage efficiency and data ingestion processes using Logstash.
- Demonstrated expertise in Kafka usage and CI/CD pipelines via Azure DevOps and Jenkins. Adapted well to dynamic data science project environments following agile methodologies.
- Experienced with NOSQL technologies like MongoDB and SQLite. Developed APIs using FAST API for efficient model inferencing and data sharing within the team.
- Utilized Apache Zeppelin notebook and developed user-defined functions (UDFs) in PySpark to address specific business needs.
- Leveraged Social Media Data for Topic Modeling and Twitter Sentiment Analysis, extracting valuable insights from Twitter data.
- Hands-on experience with Hugging Face Transformer models, including BERT.
- Conducted advanced Twitter analysis using state-of-the-art BERT models for sentiment analysis and LDA for topic modeling.
- Visualized sentiment and topic analysis outcomes using interactive dashboards crafted with Plotly and Power BI.
- Developed dashboards that provided clear insights into Twitter sentiments and trending topics, enhancing engagement and decision-making.

Technologies and Languages

- **Languages:** Python, PySpark
- **Generative AI :** MCP, Multiagentic framework, GPT vision model, Llama3.1, GPT 3.5, GPT 4, Microsoft Phi-2, Llama2, Falcon LLMs
- **Prompt Enginerring – (COD)** Chain of density, Few shot Prompting, ReAct Prompting
- **Machine Learning:** Boosting, Clustering, LDA, Time series- Prophet, ARIMA models, BERT, YOLO-NAS, Plotly,Sns
- **Tools:** Azure Devops CI CD pipline, AWS, Databricks GCP, Jenkins, Elastic search, Logstash, Kibana, MLflow, Airflow, Docker
- **Database:** MySQL, Mongo dB, SQL Alchemy, PostgreSQL
- **API :** Flask, FastAPI
- **Others:** Power BI, LangGraph, Langchain, Llamaindex, Google OCR, pytesseract, Transformers model, Deep Learning models, NLP, Computer vision, Langchain, Pytorch, QLORA, PEFT, Semantic Search,Vector databases – Pinecone DB, Chroma DB

Education and Certifications

- **Master of engineering,** JSPM institute, Pune. **2016–2018**
- **Advanced Certification in AI ML,** IIIT, Hyderabad. **2019–2020**
- **(AI-100)** Microsoft Certified : AI Fundamentals **2021**
- **(DP-100)** Microsoft Certified : Data Scientist Associate **2022**
- **(AI 102)** Microsoft Certified : Azure AI Engineer Associate **2025**

Achievements and Awards

- Eminence Award **Jan 2024**
- Pinnacle Award **July 2022**
- Rookie **Jun 2021**